

Practice problems #10

1. Let $A = \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & d_n \end{bmatrix}$. Calculate e^{At} .

2. Let $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$.

- (a) Show that $A^2 = -I$. This shows the insight of how imaginary unit i came in place that square of something could be a "negative" thing, that is,

$$i^2 = -1.$$

- (b) Calculate e^{At} .

- (c) Show by actual calculation that

$$\frac{d}{dt}e^{At} = Ae^{At}.$$

Answers or Hints are on the next page.

$$\text{\#1: } e^{At} = \begin{bmatrix} e^{d_1 t} & 0 & \dots & 0 \\ 0 & e^{d_2 t} & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & e^{d_n t} \end{bmatrix}.$$

\#2(a): Show!

$$\text{\#2(b): } e^{At} = \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix}. \quad \text{\#2(c): Do it!}$$